MIL-G-62061B(AT)
26 April 1988
SUPERSEDING
MIL-G-62061A(AT)
14 December 1973

MILITARY SPECIFICATION

GENERATOR, ENGINE ACCESSORY: 28 VOLT DC,
RATED OUTPUT 300 AMPERE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 28 volt (v) direct current (dc), blower-cooled generator with rated output of 300 amperes (A) (see 6.1).

1.2 Classification. The generators are classified based on differences in their configurations as follows:

Configuration A	- Generator (Drawing No. 871742) with mechanically driven blower.
Configuration B	- Generator (Drawing No. 10889713) with built in electrically driven blower.
Configuration C	- Generator (Drawing No. 10889998) with mounted electrically driven blower.
Configuration D	- Generator (Drawing 10914831).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2920

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.
TT-P-1757	- Primer Coating, Zinc Chromate, Low Moisture Sensitivity.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-11693	- Capacitors, Feed Through, Radio-Interference Reduction, AC and DC (Hermerically Sealed in Metal Cases), Established and Non-Established Reliability, General Specification for.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354.
MIL-I-45208	- Inspection System Requirements.
MIL-C-81562	- Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited).

STANDARDS
FEDERAL

FED-STD-H28/2	- Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting, Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461A	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.
MS51004	- Generator, Engine Accessory: 28 Volts, DC, 300 Ampere.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

8717421	- Generator, Engine Accessory.
10882774	- Cradle, Generator Engine Accessory Assembly.
10884044	- Adapter.
10889713	- Generator, Engine Accessory.
10898759	- Fan, Electrical Rotating Equipment.
10898795	- Generator, Engine Accessory.
10912456	- Brushes, Generator 300 AMPS.
10889998	- Generator, Engine Accessory.
10914831	- Generator, Engine Accessory (M107).

(Copies of specifications, standards, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in

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the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633 - Electrodeposited Coatings of Zinc on Iron and Steel, Specification for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish the first completed generator which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent generators delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and drawings. Materials shall be uniform and free of defects and imperfections that might affect the serviceability and appearance of the finished product (see 4.8.1).

3.2.1 Dissimilar metals. Dissimilar metals as defined by MIL-STD-889 shall not be used in contact with each other. Where such contact is unavoidable, the metal shall be protected against electrolytic corrosion. When protection is used, it shall be of such type that a low impedance path is offered to radio frequency currents (see 4.8.1).

3.2.2 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.8.1 and 6.4.1).

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3.3 Design and construction. Generator assemblies shall conform to Drawing 10898795 and to the applicable configuration specified below (see 1.2 and 4.8.2).

Configuration A	- Generator (Drawing No. 871742) with mechanically driven blower.
Configuration B	- Generator (Drawing No. 10889713) with built in electrically driven blower.
Configuration C	- Generator (Drawing No. 10889998) with mounted electrically driven blower.
Configuration D	- Generator (Drawing 10914831).

3.3.1 Interchangeability. All generator assemblies and all component parts shall be physically and functionally interchangeable with all other component parts of same part number previously furnished under this specification (see 4.8.2).

3.3.2 Standard parts. Military standard parts shall be used whenever possible (see 4.8.2).

3.3.3 Threaded parts. Screw threads of the form, number per inch and class specified on the applicable drawing or military standard shall be in accordance with FED-STD-H28/2 (see 4.8.1 and 4.8.2).

3.3.4 Polarity. The generator shall be designed to operate within a negatively grounded electrical circuit (see 4.8.1 and 4.8.2).

3.3.5 Bearings. Bearings included in generator assemblies or supplied as repair parts shall meet the requirement of FF-B-171 together with any additional requirements specified on applicable standards or drawings. Bearing lubrication shall conform to MIL-G-23827 (see 4.8.1 and 4.8.2).

3.3.6 Capacitors. Capacitors included in generator assemblies or supplied as repair parts shall meet the requirements of MIL-C-11693 (see 4.8.1 and 4.8.2).

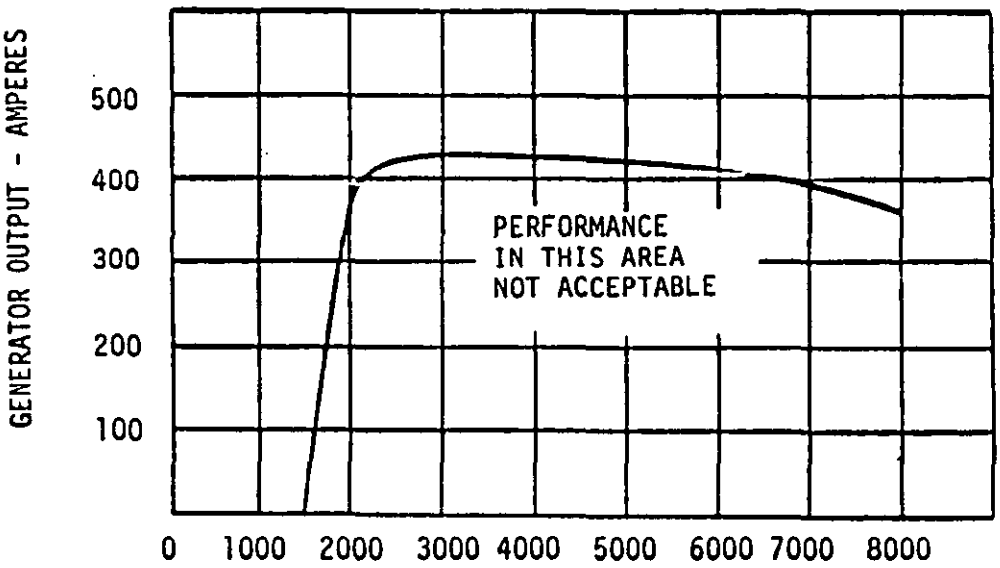
3.3.7 Brushes. Brushes included in generator assemblies or supplied as repair parts shall conform to Drawing 10912456 (see 4.8.1 and 4.8.2).

3.4 Performance.

3.4.1 Output. When tested as specified in 4.8.3, the generator shall meet the performance requirements specified in figure 1 and MS51004.

3.4.1.1 Output, performance checkpoint (heat stabilized). When tested as specified in 4.8.3.1, the generator shall subsequently meet the requirements at the one stabilized performance checkpoint indicated in figure 1.

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RPM	AMP	
1500 -	0	- CUT-IN SPEED
1625 -	100	
1750 -	200	
1900 -	300	- MIN SPEED AT RATED OUTPUT
2000 -	380	- PERFORMANCE CHECK POINT
2500 -	425	
3000 -	425	
4000 -	425	
5000 -	425	
6000 -	415	
8000 -	360	- MAX OPERATING SPEED
10000 -		- OVERSPEED

FIGURE 1. Performance curve.

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3.4.1.2 Speeds and current. When tested as specified in 4.8.3.2, the generator shall meet the current and voltage requirements at less than the maximum speeds specified.

3.4.2 Overspeed. When tested as specified in 4.8.4, the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, raised commutator bars or other damage. The generator shall subsequently meet the requirements of 3.4.1.1.

3.4.3 Interpole voltage. When the generator is tested as specified in 4.8.5, the voltage measured between the paralleling terminal (D) and the negative terminal (E) shall conform to limits shown on figure 2.

3.4.4 Compounding. When the generator is tested as specified in 4.8.6, the field current shall always increase when the load is increased.

3.4.5 Insulation resistance. When the generator is tested as specified in 4.8.7, it shall have a resistance of not less than 25 megohms, conductor to conductor and conductor to ground.

3.4.6 Ripple. When the generator is tested as specified in 4.8.8, it shall meet the ripple requirements of MIL-STD-1275.

3.4.7 Dielectric strength. When the generator is tested as specified in 4.8.9, the insulation shall evidence no loosening, cracking, charring, burning, smoking or reduction of dielectric strength.

3.4.8 Environmental.

3.4.8.1 High temperature. When tested as specified in 4.8.10.1, the generator frame temperature shall not exceed 320 degrees Fahrenheit (°F). The generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, raised commutator bars or other damage. The generator shall subsequently meet the requirements of 3.4.1.1.

3.4.8.2 Low temperature. When tested as specified in 4.8.10.2, the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, raised communtator bars or other damage. The generator shall subsequently meet the requirements of 3.4.1.1.

3.4.8.3 Shock resistance. When tested as specified in 4.8.10.3, the generator shall subseqently meet the requirements of 3.4.1.1.

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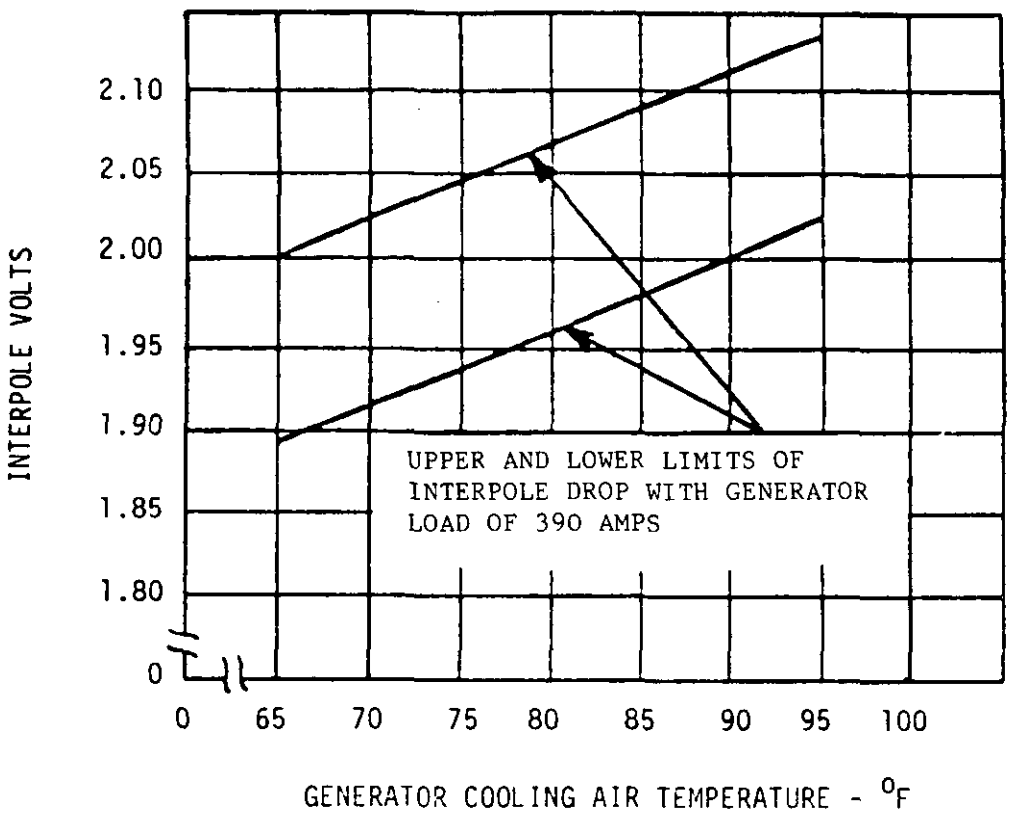


FIGURE 2. Generator interpole voltage drop at 390 Amp load vs generator cooling air temperature.

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3.4.8.4 Vibration resistance. After testing as specified in 4.8.10.4, the generator shall meet the requirements of 3.4.1.1.

3.4.8.5 Fungus resistance. After testing as specified in 4.8.10.5, the generator shall meet the requirements of 3.4.1.1.

3.4.8.6 Corrosion resistance. After testing as specified in 4.8.10.6, the generator shall meet the requirements of 3.4.1.1.

3.4.8.7 Sand and dust. After testing as specified in 4.8.10.7, the generator shall meet the requirements of 3.4.1.1.

3.4.9 Generator flexible drive.

3.4.9.1 Torsional vibration, 100 hours. When tested as specified in 4.8.11.1, the flexible drive shall limit the armature amplitude to within plus or minus five degrees in torsional vibration.

3.4.9.2 Torsional vibration, 15 minutes. When tested as specified in 4.8.11.2, the flexible drive shall limit the armature amplitude to within plus or minus seven degrees in torsional vibration.

3.4.10 Electromagnetic compatibility. When tested as specified in 4.8.12, the generator shall meet the requirements of MIL-STD-461A, Notice 4, Methods RE02 and CE07.

3.4.11 Endurance. When tested as specified in 4.8.13, the generator shall deliver sequentially, as shown, output for 1000 hours. During this time, it shall require no servicing or replacement of parts. The generator shall subsequently meet the requirements of 3.4.1.1.

3.5 Painting and surface finishing (see 4.8.2.

3.5.1 Painting. All exterior surfaces of the generator assembly shall be painted except the electrical connections, terminal box, and the extended shaft. No paint shall be applied to mating surfaces on the frame or on the ends of the generator.

3.5.2 Ferrous surfaces. Ferrous metal surfaces shall be cleaned in accordance with the applicable method specified in TT-C-490. The cleaned surfaces shall be treated in accordance with type I or type III of TT-C-490. Exterior surfaces shall be given one coat of primer conforming to TT-P-636, 0.75 to 1.25 mil thick, or TT-P-1757, 0.40 to 0.60 mil thick. The finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.5.3 Aluminum alloy surfaces. Aluminum surfaces shall be cleaned and treated in accordance with MIL-C-5541, or anodized in accordance with MIL-A-8625.

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3.5.4 Zinc plated surfaces. Zinc plated surfaces shall conform to ASTM B633, or type I, class 3 of MIL-C-81562.

3.5.5 Stainless steel parts and surfaces. –Stainless steel parts and surfaces conforming to military specification need not be plated or painted.

3.6 Identification and marking (see 4.8.2).

3.6.1 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.6.2 Identification. A nameplate shall be affixed to the exterior of the generator frame in a location specified on applicable drawings (see 3.3). Nameplates shall conform to the non-dimensional requirements of MIL-P-514 and shall include the following minimum information:

Generator, 28 Volts DC.
300 Amperes.
Army part number.
National stock number.
Manufacturer's identification.
Manufacturer's serial number.
US

3.6.3 Permanent mark. When specified (see 6.2), the Army part number required on the nameplate shall be permanently inscribed on the generator frame in such location as not to be covered by the nameplate.

3.6 Workmanship. The workmanship shall be such as to assure a product free of rust, burrs, and scratches. All electrical connections shall be mechanically secured before being soldered or brazed, to preclude the possibility of becoming loose due to vibration or shock. All solder cleaning agents and flux shall be noncorrosive. All traces of flux shall be chemically neutralized or removed after soldering (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions.

4.3.1 Atmospheric limits. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77 $\pm$ 15°F
b. Barometric pressure $^{725 + 50}_{- 75}$ millimeter (mm) mercury
c. Relative humidity 50 $\pm$ 30%

4.3.2 Laboratory test conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. The ambient room temperature shall be maintained at 77 $\pm$ 15°F. The test units shall be stabilized for 1 hour before being subjected to the tests. The generator shall be tested in combination with a blower conforming to Drawing 10898759, using adapter conforming to Drawing 10884044 assembled in accordance with Drawing 10889713 (see 6.5). The test circuit of figure 3 or Government approved equivalent shall be used whenever applicable. Readings shall be corrected for variations in the ambient temperature by use of the following formula:

$$\text{Output (corrected)} = \frac{\text{Output (observed)}}{1 + .00214\ (77-T)}$$

Where T = ambient temperature, °F

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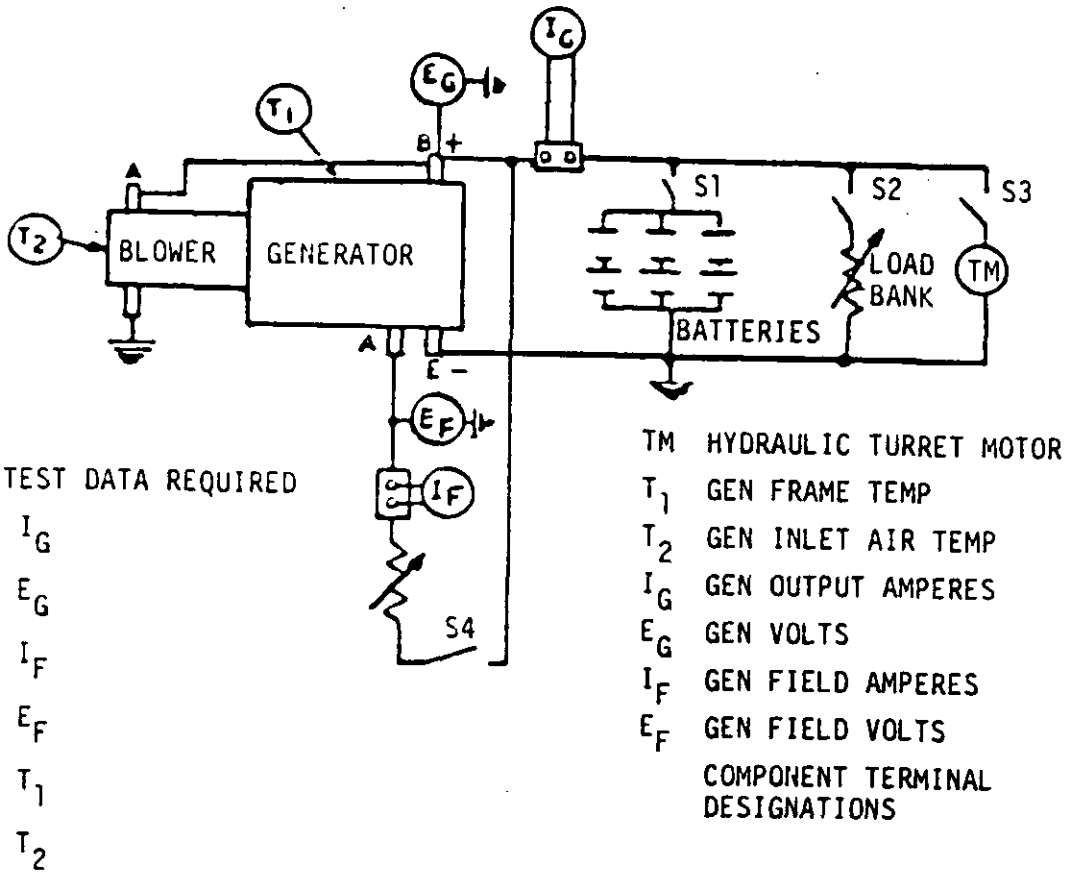


FIGURE 3. Generator test circuit.

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4.3.3 Apparatus. In addition to standard measuring equipment (see table I, figure 3 and applicable test specifications), test apparatus shall include chambers for conducting low temperature (minus 80°F), high temperature (plus 200°F), sand and dust tests, and a submersion tank.

4.3.4 Calibration. Test equipment accuracy shall be such as to permit measurement of 10% of product or test specification tolerance. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to conduct an inspection, where accuracy of test equipment has not been established to the satisfaction of the Government. Standard measuring instrument tolerance accuracies are listed in tables I and II.

TABLE I. Measuring instrument tolerances for qualification.

Measuring instrument	Accuracy
Voltmeter, dc	+ 0.25% (30 V full scale)
Ammeter, dc	± 0.50% (500 A full scale)
Tachometer	± 1%
Thermocouple system	± 2% (400°F full scale)

TABLE II. Calibration of quality conformance and control test intrumentation.

Measuring instrument	Accuracy
Voltmeter, dc	+ 0.15 V
Ammeter, dc	+ 2.50 A
Tachometer	+ 50 revolutions per minute (rpm) at 1800 rpm + 200 rpm at 10 000 rpm
Thermocouple system	+ 2% (400°F full scale)

4.3.5 Frame temperature. The frame temperature shall be measured by means of a thermocouple inserted in a hole drilled in or adjacent to the pole screw nearest the cooling air outlet of the generator. The poleshoe screw shall not be loosened.

4.3.6 Stabilized temperature. The generator system shall be considered to be temperature stabilized when operation under constant conditions fails to cause a change in frame temperature in excess of 2°F in any 5-minute period of any 20-minute test.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select one generator produced under the production contract for first article inspection. First article samples shall be inspected as specified in 4.5.2 and tested as specified in tables III and IV. Approval of the first article sample by the Government shall not relieve the contractor

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of his obligation to supply generators that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies found during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Materials and construction	3.2 thru 3.3, 3.3.3 thru 3.3.7	4.8.1	X			
Defects (see 4.5.2 and table V)	3.3 thru 3.3.7, 3.5 and 3.6	4.8.2	X	X		
Output	3.4.1	4.8.3	X			X
Performance checkpoint	3.4.1.1	4.8.3.1	X			
Speeds and current	3.4.1.2	4.8.3.2	X		X	
Overspeed	3.4.2	4.8.4	X			X
Interpole voltage	3.4.3	4.8.5	X			
Compounding	3.4.4	4.8.6	X			
Insulation resistance	3.4.5	4.8.7	X		X	X
Ripple	3.4.6	4.8.8	X			
Dielectric strength	3.4.7	4.8.9	X			
High temperature	3.4.8.1	4.8.10.1	X			
Low temperature	3.4.8.2	4.8.10.2	X			
Shock resistance	3.4.8.3	4.8.10.3	X			
Vibration resistance	3.4.8.4	4.8.10.4	X			

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TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Fungus resistance	3.4.8.5	4.8.10.5	X			
Corrosion resistance	3.4.8.6	4.8.10.6	X			
Sand and dust	3.4.8.7	4.8.10.7	X			
Torsional vibration, 100 hours	3.4.9.1	4.8.11.1	X			
Torsional vibration, 15 minutes	3.4.9.2	4.8.11.2	X			
Electromagnetic compatibility	3.4.10	4.8.12	X			
Endurance	3.4.11	4.8.13	X			X

TABLE IV. Order of first article testing.

Samples		Test
1	4.8.9	Dielectric strength
	4.8.3	Output
	4.8.4	Overspeed
	4.8.10.1	High temperature
	4.8.10.2	Low temperature
	4.8.10.4	Vibration
	4.8.10.6	Corrosion resistance
	4.8.10.7	Sand and dust
2	4.8.3	Output
	4.8.12	Electromagnetic compatibility
	4.8.11	Generator drive (torsional vibration)
3	4.8.9	Dielectric strength
	4.8.3	Output
	4.8.6	Compounding
	4.8.5	Interpole voltage
	4.8.4	Overspeed
	4.8.10.1	High temprature
	4.8.7	Insulation resistance
	4.8.10.2	Low temperature

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TABLE IV. Order of first article testing - Continued.

Samples		Test
	4.8.10.3 4.8.10.5 4.8.8	Shock resistance Fungus resistance Ripple
4	4.8.9 4.8.3 4.8.4 4.8.10.1 4.8.10.2 4.8.13 4.8.8	Dielectric strength Output Overspeed High temperature Low temperature Endurance Ripple

4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the generators of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for quality conformance inspection.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each generator selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table V.

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TABLE V. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly incomplete (see 3.3).	Visual and functional
102	Dimensions affecting interchangeability or safety out of tolerance (see 3.3.1 and 3.3.2).	SIE 1/
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability out of tolerance (see 3.3.1 and 3.3.2).	SIE
202	Improper finish (see 3.5).	Visual
203	Improper marking (see 3.6.1).	Visual
204	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests (100%). Each generator shall be subjected to tests as specified in table III with the insulation resistance test, 4.8.7, conducted first. Generator frame temperatures are not stabilized for these tests.

4.5.4 Rejection and retest. Generators that have been rejected may be reworked or have parts replaced to correct defects and be resubmitted for tests. Before resubmitting, detailed information or data concerning the rejection and the action taken to correct the defects shall be furnished to the Governmental inspector.

4.6 Control tests.

4.6.1 Sampling for control tests. Control test samples shall be selected at the rate of two generators for every 250 produced (or portion therof), except that not more than four generators shall be selected in any 30-day period. Samples shall be from a production lot which has passed the quality conformance examination specified in 4.5.2 and quality conformance tests 4.5.3.

4.6.2 Control test procedures. Control test samples shall be identified as to production period and subjected to the tests specified in table VI.

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TABLE VI. Control test sequence.

	Requirement	Test
3.4.5	Insulation resistance	4.8.7
3.4.1	Output	4.8.3
3.4.2	Overspeed	4.8.4
3.4.11	Endurance (reduced) 1/	4.8.13

1/ Test steps of 45 minutes duration.

4.7 Failure. Failure of any generator to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3, and 3.3.3 through 3.3.7 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3 through 3.3.7, 3.5 and 3.6 shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Output. To determine conformance to 3.4.1, the generator shall be stabilized at the highest attainable frame temperature, not exceeding 165°F above ambient temperature, at 28.5 V. The generator shall be stabilized at this temperature, unless some factor other than temperature, such as poor commutation, limits the performance at the particular point. To determine conformance, at least six points shall be established on the performance curve, including the following as applicable:

a. Cut-in speed (see 6.4.3). This speed shall be determined with no resistance in series with the field.
b. Minimum speed at rated output (see 6.4.5). This speed shall be determined with no resistance in series with the field.
c. Maximum operating speed (see 6.4.4). This shall be the maximum operating speed as shown on the drawing. It shall be established by adjusting the field resistance.
d. Performance check point (see 6.4.6).

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4.8.3.1 Output, performance checkpoint (heat stabilized). To determine conformance to 3.4.1.1, the generator shall be subjected to the same test procedure specified in 4.8.3 except that the output shall be determined at only the one performance checkpoint indicated on the performance curve on figure 1.

4.8.3.2 Speeds and current. To determine conformance to 3.4.1.2, the armature shall rotate freely except for brush drag. Each generator shall be tested at room temperature, 77 ± 15°F and shall not exceed the following maximum speeds:

a. Cut-in speed at 28.5 ± 0.1 V at 0 A load
(Maximum speed allowable: 1425rpm)
b. Minimum speed at 100 A at 28.5 ± 0.1 V
(Maximum speed allowable: 1550 rpm)
c. Minimum speed at 300 A at 28.5 ± 0.1 V
(Maximum speed allowable: 1825 rpm)

4.8.4 Overspeed. The generator shall be stabilized at maximum operating speed. The field circuit shall then be opened and generator speed increased to the overspeed specified on the applicable drawing for 1 minute. The generator shall then be inspected for damage to determine conformance to 3.4.2. Subsequent to the test the generator shall be subjected to the test specified in 4.8.3.1.

4.8.5 Interpole voltage.

4.8.5.1 Instrumentation. A D'Arsonval type movement meter shall be used for voltage and current measurements during tests. A mercury-bulb thermometer shall be used for temperature measurments of generator cooling fan intake air. Locate thermometer in line with fan centerline at a distance of 3 foot-inches ± 6 foot-inches from fan. Shield thermometer from any hot surface radiations that may be present. The generator inlet cooling air shall be ducted to simulate the ducting configuration of the generator installation of the M60A1 vehicle (see 3.4.3).

4.8.5.2 Test operation. To determine conformance to 3.4.3, the generator shall be operated with a separately excited field (from a constant power supply) at 5500 rpm with a resistive load of 390 A at 28 V until temperature conditions stabilize. After stabilization, record interpole voltage, load current, terminal voltage and cooling-air temperature. Using figure 2, find the cooling air temperature equivalent to that recorded. The recorded interpole voltage of the generator must fall within the limits (figure 2) on the ordinate determined by the cooling air temperature.

4.8.6 Compounding. Generators shall be operated at 6500 rpm and the output voltage adjusted and maintained constant at 28.5 ± 0.1 volts while zero, half and full-rated loads are successively placed on the generator. Field current shall be measured to determine conformance to 3.4.4.

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4.8.7 Insulation resistance. To determine conformance to 3.4.5, the generator shall be subjected to the condition B, method 302 insulation resistance test specified in MIL-STD-202. During the test the generator and motor commutator brushes shall be raised off the commutator.

4.8.8 Ripple. To determine conformance to 3.4.6, the generator shall be operated in the test circuit with the battery disconnected. Ripple voltage shall be observed and recorded under the conditions listed in table VII.

TABLE VII. Ripple voltage.

Generator rpm	Generator V ± 0.2 V	Generator Output A
2400	28	0
2400	28	100
2400	28	400
4000	28	0
4000	28	100
4000	28	400
6000	28	0
6000	28	100
6000	28	400
8000	28	0
8000	28	100
8000	28	375

4.8.9 Dielectric strength. To determine conformance to 3.4.7, the generator shall be subjected to a ground test of 500 V root mean square (rms) at a frequency of 60 cycles per second (cps) applied for 1 minute. The potential shall be applied to the armature and field terminals with the ground brush raised off the commutator, and with the field ground terminal disconnected.

4.8.10 Environmental.

4.8.10.1 High temperature. The generator shall be stored for 24 hours in an ambient temperature of 160 ± 5°F. The generator shall then be stabilized at average speed at rated output at ambient temperature of 160 ± 5°F. The generator frame temperature shall be recorded. After stabilization the generator shall be operated under these conditions for 4 hours. The generator shall then be inspected to determine conformance to 3.4.8.1. Subsequent to the test the generator shall be subjected to the test specified in 4.8.3.1.

4.8.10.2 Low temperature. The generator shall be stored for 24 hours in an ambient temperature of minus 65 ± 5°F. The generator shall then be stabilized at average speed at rated output at an ambient temperature of

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minus 65 ± 5°F. After stabilization, the generator shall be operated under these conditions for 4 hours. The generator shall then be examined to determine conformance to 3.4.8.2. Subsequent to the test, the generator shall be subjected to the test specified in 4.8.3.1.

4.8.10.3 Shock resistance. To determine conformance to 3.4.8.3, the generator shall be tested as specified under condition A, method 213 medium impact shock test of MIL-STD-202. During the test the generator shall be mounted on a cradle conforming to Drawing 10882774. Subsequently, the generator shall be subjected to the test specified in 4.8.3.1.

4.8.10.4 Vibration resistance. To determine conformance to 3.4.8.4, the generator shall be subjected to the method 201 vibration test specified in MIL-STD-202, except that the test shall be repeated 12 times for a total duration of 72 hours. During the test, the generator shall be operated at a speed of 4,500 ± 500 rpm. During the test, the generator shall be mounted on a cradle conforming to Drawing 10882774. Subsequently the generator shall be subjected to the test specified in 4.8.3.1.

4.8.10.5 Fungus resistance. To determine conformance to 3.4.8.5, the generator shall be tested in accordance with class 2, method B of MIL-F-13927 except that the test period shall be a continuous 90 days. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without disturbing any fungus growths that have developed. Upon completion of the examination and replacement of access port covers, the generator shall be operated for 1 hour, without load at 6500 ± 500 rpm before being subjected to the testing specified in 4.8.3.1.

4.8.10.6 Corrosion resistance. To determine conformance to 3.4.8.6, the generator shall be tested for resistance to corrosion in accordance with test condition A of method 101 of MIL-STD-202, except that the length of the test shall be 200 hours. Prior to the performance tests, the generator shall be operated for 1 hour at 6500 ± 500 rpm without load.

4.8.10.7 Sand and dust. To determine conformance to 3.4.8.7, the generator shall be subjected to test condition A of method 110 of MIL-STD-202 except that air speed shall be 2500 ± 500 feet per minute.

4.8.11 Generator drive. To determine conformance to 3.4.9, the generator shall be tested as specified in 4.8.11.1 and 4.8.11.2. The test shall be conducted on a universal joint torsional vibration machine which has a flywheel with at least 20 times the moment of inertia of the armature being tested.

4.8.11.1 Torsional vibration, 100 hours. The armature shall be subjected to a 100-hour torsional vibration test with plus or minus one degree torsional amplitude input to the drive shaft at critical frequencies.

4.8.11.2 Torsional vibration, 15 minutes. The armature shall be subjected to a 15-minute torsional vibration test with plus or minus two degrees torsional amplitude input to the drive shaft at critical frequencies.

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4.8.12 Electromagnetic compatibility. To determine conformance to 3.4.10, the generator shall be tested in accordance with of MIL-STD-462, Notice 3.

4.8.13 Endurance. To determine conformance to 3.4.11, the generator shall be operated as detailed below:

TABLE VIII. Endurance test cycles.

Test Step	Generator, rpm (± 25 rpm)	Generator, V(± 0.2 V)	Generator, Output, A (minimum)	Time, Hours
1	2500	28.5	100	8
2	5500	28.5	200	8
3	8000	28.5	300	8
4	2500	28.5	200	8
5	5500	28.5	400	8
6	8000	28.5	100	8
7	2500	28.5	300	8
8	5500	28.5	100	8
9	8000	28.5	200	8

This cycle is repeated thru four cycles at which time the generator shall then be operated with the armature shaft at an angle of 45 ± 1 degree above the horizontal for two cycles. Return to the horizontal for four cycles, then operate the generator with the armature shaft at an angle of 45 ± 1 degree below the horizontal for two cycles. Return to the horizontal for the last two cycles to complete 1000 hours. Output current and voltage, and generator speed and temperature shall be measured during the test. Subsequent to the test the generator shall be subjected to the test specified in 4.8.3.1.

4.9 Inspection of preparation for delivery.

4.9.1 Materials and processes. The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5 and specifications referenced herein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The generator covered by this specification is used in combination with a mechanically driven blower on the early M48 series, M551 and M551A1 vehicles. The generator in combination with an electrically driven blower is used on the M48A5, M60A1, M88A1, M578 and M110 tracked vehicles. Differences in the various installation drawings result from varying ducting configurations. The M60A1 installation is used in the specification as the test arrangement for evaluating the basic generator. When applicable, the installation assemblies are required to conform to a pressure leakage test, as specified on the respective drawings.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. If first article samples are not required (see 3.1).
c. If permanent marking shall be inscribed on the frame (see 3.6.3).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If atmospheric limits are other than as specified (see 4.3.1).
g. If the number of first article samples shall be other than as specified (see 4.4).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 Special equipment.

6.3.1 Transient characteristics. CEC oscillographic tape has furnished satisfactory results for measuring transient characteristics.

6.3.2 Voltage ripple. Tektronix scope, type 545 has proven adequate for measuring voltage ripple.

6.4 Definitions.

6.4.1 Recovered materials. Recovered materials means materials that have been collected or recovered from solid waste (see 6.4.2 and 3.2.2).

6.4.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in

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domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4.3 Cut-in speed. Cut-in speed is the lowest speed at which the generator, with frame temperature stabilized, will generate 28.5 V with no external load (see 4.8.3).

6.4.4 Maximum operating speed. Maximum operating speed shall be as indicated on figure 1.

6.4.5 Minimum speed at rated output. Minimum speed at rated output is the lowest speed at which rated output can be obtained at 28.5 V (see 4.8.3).

6.4.6 Performance check point. The performance check point is that point on the speed output curve used as a reference point to check generator output, as indicated on figure 1.

6.4.7 Average speed at rated output. Average speed at the rated output is the speed half way between the "minimum speed at rated output" and the "maximum operating speed at rated output" at 28.5V (see 4.8.10.2).

6.4.8 Maximum operating speed at rated output. Maximum operating speed at rated output is the highest speed at which rated output is generated at 28.5 V (see 6.4.7).

6.5 Government furnished property. When required, the Government shll furnish blower, part number 10898759 and adapter, part number 10884044 (see 4.3.2).

6.6 Subject term (key word) listing.

Accessory, engine generator, 28 volt dc, 300 ampere.
DC generator, 28 volt, 300 ampere.
Engine accessory, generator, 28 volt dc, 300 ampere.
Engine generator, accessory, 28 volt dc, 300 ampere.

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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FSC 2920

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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